NEURAL MECHANISMS OF DRUG ABUSE IN FEMALES

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INTRODUCTION

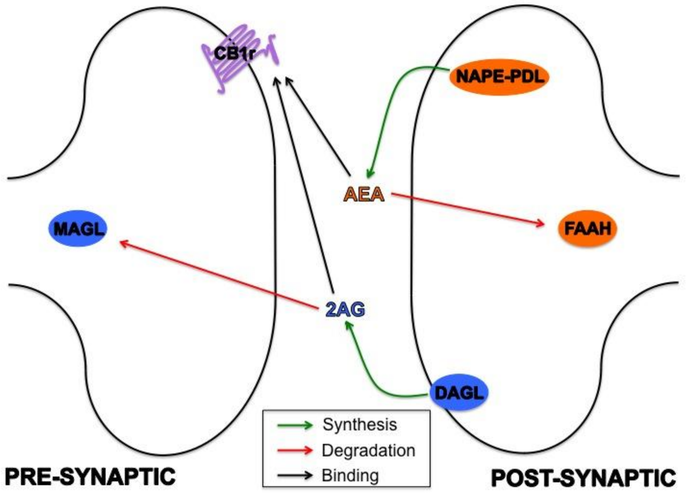
Sex differences in addiction

- Women display markedly faster progressions from casual drug use to drug abuse and addiction relative to men
- Female sex steroid, estradiol, appears to mediate this effect

Neural mechanisms of estradiol

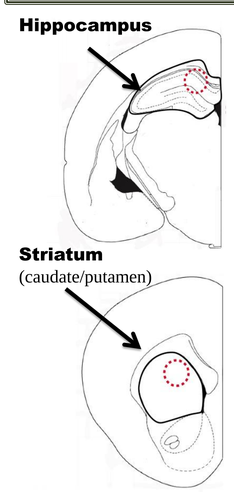
- Striatum is an integral brain region in development and expression of addictive behaviors
- Estradiol impacts striatal function
- Endocannabinoid system may mediate some effects of estradiol

Does the endocannabinoid system mediate the effects of estradiol within the striatum to drive addictive behavior in females?



We hypothesized that estradiol mediates the expression of molecules that synthesize, degrade, and bind the two best studied endocannabinoids in the striatum, to ultimately drive addictive behaviors in females.

METHODOLOGY



Start

- Subcutaneous injection of cottonseed oil ($n = 4$) or estradiol (10µg; $n = 5$) in OVX female Sprague-Dawley rats

24 hrs

- Sacrifice
- Tissue punches of caudate/putamen, and hippocampus

48 hrs

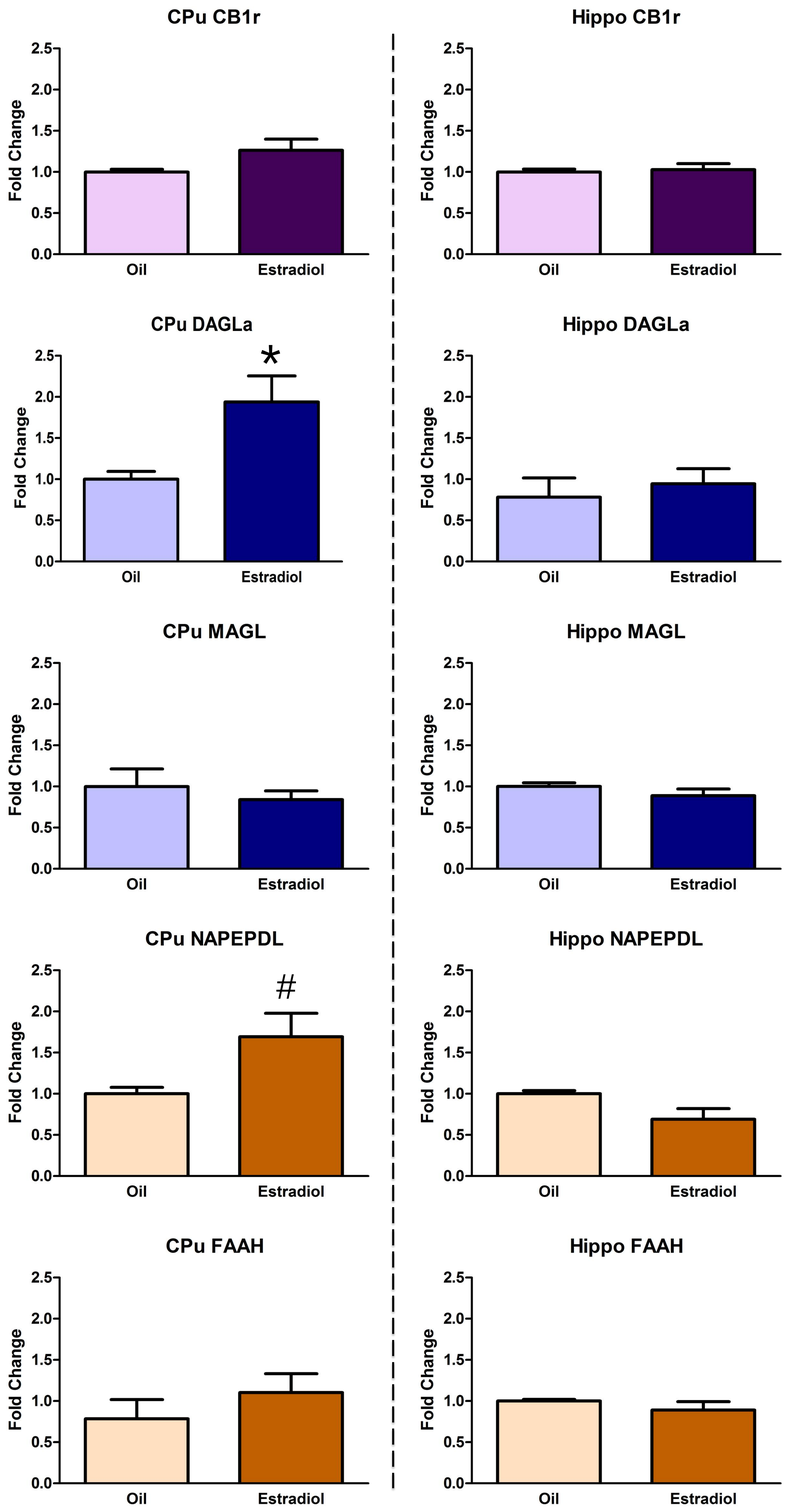
- Extract RNA (Qiagen kit)
- Reverse transcribe to cDNA (Roche cDNA synthesis kit)

72 hrs

- qRT PCR (Roche LightCycler 480)
- Analyze stats (independent sample t-tests)

End

RESULTS

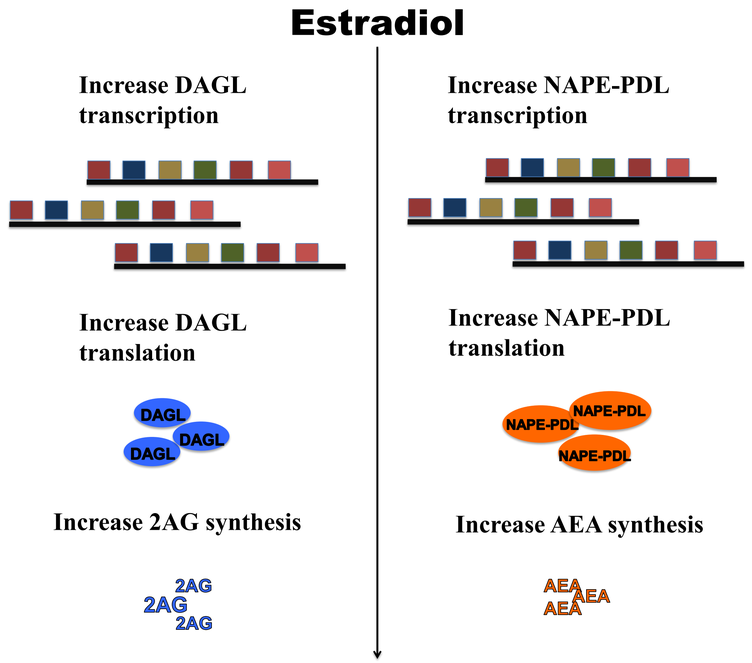


In the striatum, estradiol treatment increases expression of endocannabinoid synthesizing enzymes DAGL and NAPE-PDL, but does not affect expression of endocannabinoid degrading enzymes (MAGL, FAAH) or endocannabinoid receptor CB1. In the hippocampus, a region not integral to addiction, estradiol does not mediate any examined components of the endocannabinoid system. *$p<0.05$; #$p=0.065$.

CONCLUSIONS

- **Estradiol enhances the expression of molecules that synthesize endocannabinoids in the striatum**
- **Estradiol may not affect the expression of molecules that degrade and/or bind endocannabinoids in the striatum or the hippocampus**
- **Estradiol may increase endocannabinoid levels within the striatum and this may play a role in female responses to drugs**

MODEL



FUTURE DIRECTIONS

- Directly measure endocannbinoid levels within striatum
- Investigate estrogenic mediation of endocannabinoid system in nucleus accumbens (another brain region within striatum integral to development and expression of addictive behaviors)
- Determine whether this action of estradiol is genomic, non-genomic or both
- Investigate endocannabinoid synthesis, degradation and binding in male brain

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